

# Python

Course Duration: 20Hrs

| S. No | Content                                                                                                                                                                                                                          |
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| 1     | <b>Introduction of python</b> <ul style="list-style-type: none"><li>• What is python</li><li>• Who Developed python</li><li>• Features of python</li><li>• Applications of python</li><li>• Setting up the Environment</li></ul> |
| 2     | <b>Python Basic Syntax and Data Types</b> <ul style="list-style-type: none"><li>• Input/output</li><li>• Comments</li><li>• Variables</li><li>• Data types</li><li>• Typecasting</li></ul>                                       |
| 3.    | <b>Operators in Python</b> <ul style="list-style-type: none"><li>• Arithmetic</li><li>• Assignment</li><li>• Comparison</li><li>• Logical</li><li>• Identity</li><li>• Membership</li></ul>                                      |

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| 4. | <b>Strings in Python</b> <ul style="list-style-type: none"><li>• Creating</li><li>• Formatting</li><li>• Indexing</li><li>• Slicing</li><li>• String methods</li></ul>                  |
| 5. | <b>Lists in Python</b> <ul style="list-style-type: none"><li>• Creating</li><li>• Properties</li><li>• Indexing</li><li>• Slicing</li><li>• Methods</li><li>• Modifying lists</li></ul> |
| 6. | <b>Tuples in Python</b> <ul style="list-style-type: none"><li>• Syntax</li><li>• Properties</li><li>• Indexing</li><li>• Slicing</li><li>• Methods</li></ul>                            |

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| 7. | <b>Sets in Python</b> <ul style="list-style-type: none"><li>• Syntax</li><li>• Updating</li><li>• Operations</li><li>• Methods</li></ul>                                |
| 8. | <b>Dictionaries in Python</b> <ul style="list-style-type: none"><li>• Syntax</li><li>• Keys/values</li><li>• Accessing</li><li>• Methods</li></ul>                      |
| 9  | <b>Python Conditional Statements</b> <ul style="list-style-type: none"><li>• If statement</li><li>• if-else</li><li>• if-elif-else</li><li>• Example Programs</li></ul> |
| 10 | <b>Loops in Python</b> <ul style="list-style-type: none"><li>• while</li><li>• for</li><li>• break</li><li>• continue</li><li>• pass</li><li>• range</li></ul>          |

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| 11. | <b>Functions in Python</b> <ul style="list-style-type: none"><li>• Creating</li><li>• Calling</li><li>• Arguments</li><li>• Variables</li><li>• Recursion</li><li>• Anonymous Functions (Lambda )</li></ul> |
| 12  | <b>Modules in Python</b> <ul style="list-style-type: none"><li>• Creating</li><li>• Importing</li><li>• Built-in modules</li></ul>                                                                          |
| 13  | <b>Classes &amp; Objects</b> <ul style="list-style-type: none"><li>• Creating</li><li>• Attributes</li><li>• Methods</li><li>• Inheritance</li><li>• Polymorphism</li></ul>                                 |
| 14  | <b>Date &amp; Time</b> <ul style="list-style-type: none"><li>• date</li><li>• time</li><li>• datetime modules</li><li>• Calendar Module</li></ul> <b>Exercise &amp; Projects.</b>                           |